RAJALAKSHMI ENGINEERING COLLEGE

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P2

Started on	Thursday, 17 September 2026, 7:34 PM
State	Finished
Completed on	Thursday, 17 September 2026, 7:34 PM
Time taken	30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void solve() {
    int n1, n2;

    // Read first array
    if (scanf("%d", &n1) != 1) return;
    int a[n1];
    for (int i = 0; i < n1; i++) scanf("%d", &a[i]);

    // Read second array
    if (scanf("%d", &n2) != 1) return;
    int b[n2];
    for (int i = 0; i < n2; i++) scanf("%d", &b[i]);

    int i = 0, j = 0, first = 1;

    // Two-pointer intersection traversal
    while (i < n1 && j < n2) {
        if (a[i] < b[j]) {
            i++;
        } else if (a[i] > b[j]) {
```

```
            j++;
        } else {
            // Found a common element
            if (!first) printf(" ");
            printf("%d", a[i]);
            first = 0;
            i++;
            j++;
        }
    }
    printf("\n");
}

int main() {
    int t;
    if (scanf("%d", &t) == 1) {
        while (t--) {
            solve();
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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